The Path to a Lie*

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Abstract

This paper studies how discretionary search shapes misreporting. In an incentivized experiment embedded in the German Socio-Economic Panel Innovation Sample (SOEP-IS), 1,283 participants could open up to ten boxes containing values from 0 to 9, but were instructed to report the value in the first box they opened. Later openings could not change that value, but higher reports paid more whether or not they were truthful. Because we observe the complete click sequence, we can link what participants encounter during search to both when they stop and what they ultimately report. Participants are more likely to stop searching after finding a value that improves on everything seen before, especially when the improvement is large. When participants report above the truth after searching, their reports usually match a value encountered during search, often the highest one. Favorable values are also disproportionately likely to appear as the final opening. Together, the results reveal a close alignment between the search path and the final report: favorable values tend to mark both the end of search and the value ultimately reported. We call this *path-anchored reporting*.

Keywords Lying · Self-serving Justifications · Sequential Search · Survey

JEL Classification C91 · D82 · D90

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1 Introduction

Many economic decisions require agents to report an outcome after choosing which information to inspect and when to stop. A manager may search through performance measures until one supports a favorable review, while a researcher may examine several specifications before emphasizing the most favorable result. In these settings, additional search does not change the underlying truth, but it may make a false report easier to justify.

This argument builds on evidence that lying is psychologically costly (Abeler et al., 2019). People are less willing to misreport when a false statement feels like outright fabrication, and more willing when it connects to a defensible reference point (Mazar et al., 2008; Shalvi et al., 2011). Such justifications can arise from advantageous counterfactuals (Shalvi et al., 2011), motivated mistakes and self-serving interpretations (Exley and Kessler, 2024; Bosch-Rosa et al., 2025a), or information that individuals choose to acquire (Dimant et al., 2024; Chen and Heese, 2026).[1] Related work shows that people use avoidable information to create moral wiggle room (Dana et al., 2007; Grossman and van der Weele, 2017), distort beliefs and memory in self-serving directions (Zimmermann, 2020), and preserve flexibility in their beliefs (Saccardo and Serra-Garcia, 2023). Most of this evidence, however, comes from final choices or reports, which reveal little about how the underlying claim was produced. The same claim may be truthful, fabricated, or supported by evidence encountered only after continued search. Tracking the path to a report lets us dissect that process.

We study this process in an experiment embedded in the 2020 wave of the German Socio-Economic Panel Innovation Sample (SOEP-IS) (Richter and Schupp, 2015). Participants saw ten closed boxes containing the values 0 to 9 in random order. They were instructed to report the value in the first box they opened, but could inspect additional boxes before submitting their report. Payment equaled the reported value in euros, whether or not the report was truthful. The first opening therefore fixed the truth, while later open-

[1]The evidence on supplied counterfactuals is not settled: Clist and Hong (2026) find that allowing a second die roll does not produce a distinct increase in dishonesty. Our setting differs because participants choose whether and how far to search, so any favorable alternative value comes from their own search.

ings only changed the values participants had seen. Because the software recorded the complete opening sequence, we observe the truth, the search path, and the final report.

Our main result is that dishonest reporting is systematically related to the search path. Participants—particularly those who eventually report above the truth—are more likely to stop after an opening improves on everything they have seen, and the association grows with the size of the improvement. This is not simply a reaction to high values. For the same revealed value, participants are more likely to stop when it improves on their previous maximum. Among participants who search and later report above the truth, most choose a value they encountered during the search, few report above the highest value observed, and many report that maximum exactly. Moreover, when the final opening reveals a new high value, participants are especially likely to report a value no higher than that observed maximum. Together, these patterns reveal a close connection between the favorable values uncovered along the search path and the final report. Participants may search for evidence that makes a higher report easier to justify, or simply follow a satisficing rule. Both mechanisms imply that favorable values encountered along the way can determine where the search ends and which value is reported. We call this common behavioral pattern *path-anchored reporting*.

The paper makes three contributions. First, it isolates a setting in which additional information cannot change the truthful answer, while allowing us to observe both that answer and the participant's final report. This distinguishes the design from motivated-search settings in which information changes beliefs about the consequences of an action (Dimant et al., 2024; Chen and Heese, 2026). Here, when a participant reports falsely, we can link that report to specific values encountered only after the truthful answer was already fixed. Second, the complete search histories allow us to connect the arrival of favorable evidence to both the decision to stop and the final report, linking work on justification in dishonesty (Mazar et al., 2008; Shalvi et al., 2011; Exley and Kessler, 2024) with work on motivated information acquisition and moral wiggle room (Dana et al., 2007; Grossman and van der Weele, 2017). Third, we study these behaviors in a large household-panel sample, extending evidence on dishonesty beyond the convenience samples commonly used in experimental work (Fosgaard, 2020).

More broadly, the results suggest that distortion can arise in the process of assembling evidence, before the final report is chosen. Managers, analysts, and researchers routinely decide which outcomes to inspect and when to stop before communicating a conclusion. These choices do not alter the underlying fact, but they shape which evidence is available and salient when the claim is made. If search stops once sufficiently favorable evidence appears, part of the distortion occurs in the search process itself, even when the eventual claim rests on genuinely observed evidence. Detecting this requires observing the process—search histories, query logs, specification sequences, and stopping times—rather than the final statement alone. The box task is deliberately stylized, but it isolates this process clearly. A favorable value appears, search ends, and that value becomes an anchor for the final report.

The remainder of the paper proceeds as follows. Section 2 describes the experimental design. Section 3 presents a simple framework that motivates the empirical stopping and reporting tests. Section 4 reports the main empirical results. Section 5 concludes.

2 Experimental Design and Data

The experiment was fielded through Computer-Assisted Personal Interviews in the 2020 SOEP Innovation Sample, part of the German Socio-Economic Panel (Richter and Schupp, 2015), between September and December 2020.[2] The module was offered to 1,603 respondents; 1,330 participated, and 1,318 completed both reporting tasks. Participants completed two incentivized tasks commonly used in the experimental literature on lying, with their order randomized. In both, payment depended only on the participant's report: reporting $r \in \{0, \dots, 9\}$ paid r euros whether or not the report was truthful. Participants knew from the outset that only one task would be payoff-relevant and that a coin flip after both tasks would determine which one they were paid for. Appendix Table A.1 summarizes individual characteristics, and Appendix C reproduces the full instructions.

The main task adapts the "box paradigm" of Gneezy et al. (2018). Participants saw ten closed black boxes on the survey tablet, each containing one integer from 0 to 9 in random order. Clicking a box revealed its value. Participants could open as many boxes as they wished, but were instructed to remember and report the value in the *first* box they opened. Participants entered their report privately on the tablet, with the interviewer

[2]Despite the COVID-19 pandemic, some SOEP-IS interviews continued in person under strict hygiene protocols. For details, see the SOEP-IS fieldwork report.

instructed to leave the room or turn away. The software recorded the full sequence of clicks, including which boxes were opened, in what order, and which values were revealed. Participants were told that the laptop recorded which boxes they opened and how many, while their survey responses remained anonymous and were not linked to their names.

The raw records retain every click, including revisits to a box already opened. Because a revisit reveals no new information, from this point onward an *opening* means the first opening of a previously unseen box; unless stated otherwise, all analyses ignore revisits. Stopping is measured at the last opening before the report. Separate robustness checks either require the literal final click to open a new box or treat sequences ending in revisits as censored. The first opening identifies the instructed report; subsequent openings identify search length, observed values, and the timing of favorable updates. Of the 1,318 task completers, 1,283 opened at least one box and can therefore be classified by reporting behavior; the 392 who opened at least two boxes contribute 1,489 post-first stopping decisions.

The second task adapts the private die-rolling paradigm of Fischbacher and Föllmi-Heusi (2013). Participants rolled a fair 10-sided die under a cup as often as they liked, were asked to remember and report the *first* roll, and again entered their report privately on the tablet.[3] After both tasks, the participant visibly tossed a coin, the interviewer recorded which task was selected, and the corresponding reported number was paid in euros before the survey continued. Because the first die roll was not observed, this task identifies only aggregate excess reporting; it cannot classify individual reports as truthful or false.

Data from this module have also been used by Bosch-Rosa et al. (2025b), who study how privacy and researchers' observability of false reports affect dishonesty and social-image costs. Their analysis compares the private dice task and the observable box task, with particular attention to demographic heterogeneity. In contrast, we use click-level records from the box task to study how participants search before submitting a report, when they stop, and whether the final report is tied to values revealed along the way. Bosch-Rosa et al. (2025b) does not analyze these click-level records. Thus, while the dice task and the privacy comparison are central to Bosch-Rosa et al. (2025b), they serve here

[3] Fischbacher and Föllmi-Heusi (2013) use a 6-sided die. We use a 10-sided die to match the 0–9 support of the box task.

only as complementary benchmarks for interpreting path-based dishonesty in the box task.

3 Conceptual Framework

This section develops a simple framework for organizing the empirical analysis. The key feature of the task is that the first opening fixes the value participants are instructed to report, while later openings can only change the values they observe before deciding when to stop and what to report.

Let x_1 denote the first-box value—the truth participants are instructed to report. After $k \geq 1$ openings, let

$$m_k \equiv \max\{x_1, \ldots, x_k\}$$

denote the highest value observed so far. For $k \geq 2$, set

$$\Delta m_k \equiv \max\{0, x_k - m_{k-1}\},$$

with $\Delta m_1 \equiv 0$. The current improvement, Δm_k, is positive exactly when opening k sets a new observed maximum, and its size measures how much the new value improves on everything seen before.

If observed evidence makes a higher report easier to adopt or justify, continued search can uncover a value that supports that report. Continued search can therefore matter in two ways. A sufficiently favorable opening may bring the search to an end, while the values uncovered along the way can provide support for a higher report. We therefore expect stopping to be more likely after favorable improvements and final reports to remain closely tied to values encountered over the course of the search.

This reasoning yields two reduced-form predictions. First, among participants who are still searching, stopping should be more likely after an opening improves on the highest value seen so far, especially when the improvement is large. Second, among participants who report above the truth, reports should tend to remain within the observed range, with favorable values encountered at the end of the search especially likely to become the final report.

These predictions are consistent with more than one mechanism. A favorable opening may end the search because it makes a higher report easier to justify, or because it satisfies an aspiration level (Simon, 1955). The empirical analysis therefore focuses on whether favorable values are systematically linked to stopping and reporting, rather than on distinguishing between these mechanisms. Appendix B presents a simple reporting model that rationalizes reports both at and below the observed maximum without imposing a structural stopping rule.

4 Results

Of the 1,318 participants who completed both tasks, 1,283 opened at least one box and therefore revealed the value they were instructed to report. The remaining 35 submitted a report without opening a box.[4] Among the 1,283 participants for whom the truth is observed, 1,152 report it, 21 report below it, and 110 report above it. These 110 participants account for 8.6% of the sample and report 4.45 units above the truth on average.[5] Table 1 summarizes the sample. Search is typically short, with participants opening 2.16 boxes on average. The median participant who reports the truth opens only that box, whereas the median participant who reports above it opens three.

The relation between search and reporting is already visible in these descriptives. Of the 110 participants who report above the truth, 14 stop after the first box and 96 continue searching. Among these 96, 90 report no more than the highest value they observed. Sixty-eight report that maximum exactly, 18 report another opened value, and four report an unopened value below the maximum. Only six report above the highest value they observed. Participants who misreport without additional search report 6.0 on average, and only three choose 9, consistent with partial lying (Fischbacher and Föllmi-Heusi, 2013; Gneezy et al., 2018). Among those who search, reports are closely tied to the values encountered along the way.

[4]Their mean report is 4.89, which is not statistically different from the truthful benchmark of 4.5 (t-test, $p = 0.42$). The most common report is 5, chosen by 11 participants. Thus, those who skip the opening stage do not appear to submit unusually high reports.

[5]The 21 downward reports reduce payoffs and provide a useful benchmark for reporting error: 16 match another opened value and 19 occur after additional search. They are far less common than reports above the truth. The relatively low incidence of upward misreporting is consistent with broader evidence that many individuals forgo profitable opportunities to lie (Abeler et al., 2019).

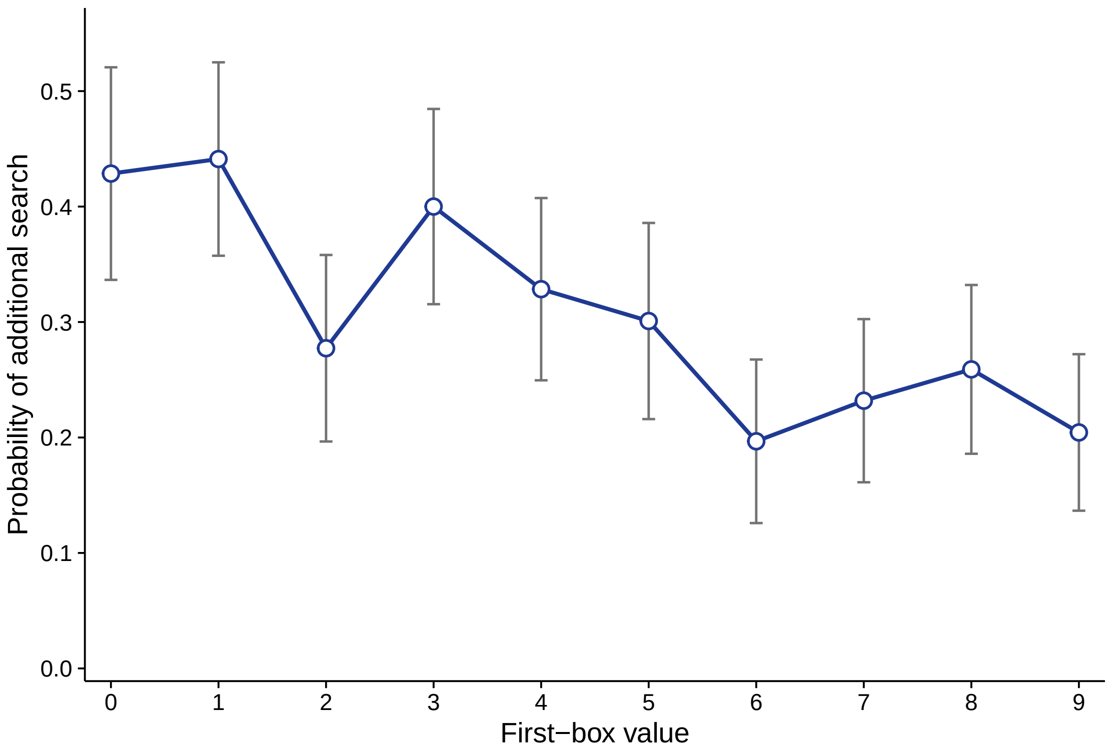
Table 1: Descriptive statistics

	Mean	SD	N
Full sample			
Reported value in box task [0–9]	4.949	2.869	1,318
First-box value	4.614	2.880	1,283
Misreport size (report - first box)	0.337	1.496	1,283
Misreported ($r > x_1$)	0.086	0.280	1,283
Boxes opened	2.161	2.273	1,283
Maximum value observed	5.825	2.868	1,283
Misreporting subset			
Boxes opened	4.000	2.726	110
Misreport size (report - first box)	4.455	2.384	110
Maximum possible gain (9 - first box)	6.591	1.988	110

Notes: The reported-value row uses the full module sample (N = 1,318). Other rows in the upper panel use valid first-box observations (N = 1,283); the lower panel uses participants who report above the first-box value (N = 110).

The random first-box value also predicts whether participants continue searching. Figure 1 shows a clear negative gradient. A one-unit increase in the first-box value lowers the probability of additional search by 2.5 percentage points. Participants whose first box contains 8 or 9 continue searching in 23.2% of cases, compared with 32.6% after values 0–7, a difference of 9.4 percentage points ($p = 0.001$; Appendix Table A.2). Searching therefore declines as the first-box value rises, with participants less likely to continue when the truth is already near the top of the payoff range.

Figure 1: Search by the randomized first-box value



Notes: Points are raw means for each first-box value among the 1,283 participants with an observed first opening; vertical bars show pointwise 95% confidence intervals, and connecting lines are visual guides. The outcome indicates whether a participant opens at least one additional box.

4.1 Favorable updates and stopping

Table 2 examines whether favorable updates predict the decision to stop searching. For each post-first opening of a previously unseen box, $Stop_{i,k}$ equals one if it is the participant's last opening of a new box before reporting and zero otherwise.[6] The key explanatory variable is *Current improvement*,

$$\Delta m_{ik} = \max\{0, x_{ik} - m_{i,k-1}\},$$

which measures how much the newly revealed value exceeds the previous maximum. It equals zero when the opening does not set a new maximum.[7]

The estimate is stable across specifications. With opening-position, current-value, prior-maximum, and first-box fixed effects, a one-unit improvement is associated with a 6.7 percentage point increase in the probability of stopping. This pattern is not simply driven by participants stopping after high values. Current-value fixed effects compare the same revealed value across different search histories, allowing us to ask whether participants are more likely to stop when that value exceeds their previous maximum. Prior-maximum fixed effects also account for differences in the highest value reached before the current opening. The remaining variation therefore captures whether crossing that previous maximum matters beyond the levels of the current and prior values. Column 4 shows that this relationship is stronger on paths that eventually lead to a report above the truth.

The second opening provides the cleanest setting for testing whether stopping responds to an improvement over the previous maximum. Among the 392 participants who open a second box, the revealed value is randomly drawn from the nine remaining values, conditional on the first-box value and on the decision to continue searching. A one-unit improvement over the truth is associated with a 7.3 percentage point increase in the probability of stopping, while simply drawing a second value above the truth raises stopping by about 20 percentage points (Table 3). Because the second value is randomized conditional on the first, this comparison isolates the effect of receiving a favorable

[6]The raw click records include occasional revisits to boxes that had already been opened. Because these clicks reveal no new information, the baseline definition ignores them and treats the last opening of a previously unseen box as the stopping point. As robustness checks, we either require the participant's literal final click to reveal a new value or treat paths ending in a revisit as censored. The results are similar under both definitions; see Appendix Table A.3.

[7]Because each observation corresponds to an opening reached by a participant who has not yet stopped, the specification can also be viewed as a discrete-time hazard model of stopping. Logit and complementary-log-log specifications yield average marginal effects of about 0.06, close to the OLS estimate; see Appendix Table A.3.

Table 2: Stopping after favorable improvements in the running maximum

	Dependent variable: *Stop*			
	(1)	(2)	(3)	(4)
Current improvement	0.060***	0.045***	0.067***	0.049***
	(0.007)	(0.009)	(0.012)	(0.013)
Misreported				-0.060**
				(0.029)
Misreported × *Current improv.*				0.052***
				(0.014)
Observations	1489	1489	1489	1489
R^2	0.140	0.155	0.175	0.184
FE	Step	Step+value	Full	Full

Notes: Opening-level OLS on post-first openings. Current improvement is $\Delta m = \max\{0, x_k - m_{k-1}\}$. The FE row reports opening-position FE, opening-position plus current-value FE, or the full set that also includes prior-maximum and first-box FE. Because current improvement is a deterministic function of current value and prior maximum, the full specification identifies its coefficient from the kink where the current value crosses the prior maximum, relative to an additive level-only stopping surface. Standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

second draw. The estimate is unchanged when we include both first- and second-value fixed effects. This specification isolates whether stopping changes when the second value exceeds the truth, rather than simply when either value is high. Appendix Figure A.2 shows the corresponding raw pattern. The response is nonlinear; improvements of one to three units have relatively modest effects on stopping, whereas estimates for improvements of four or more are substantially larger, reaching about 63 percentage points in the pooled eight-or-more bin (Figure 2). Large favorable updates therefore drive much of the relationship between improvements and stopping.

Table 3: Stopping at the second opening

	(1) Stop	(2) Stop	(3) Stop
Current improvement	0.073***		0.073***
	(0.011)		(0.021)
Second value exceeds first		0.200***	
		(0.048)	
Observations	392	392	392
R^2	0.135	0.056	0.169
First-value FE	Yes	Yes	Yes
Second-value FE	No	No	Yes

Standard errors in parentheses

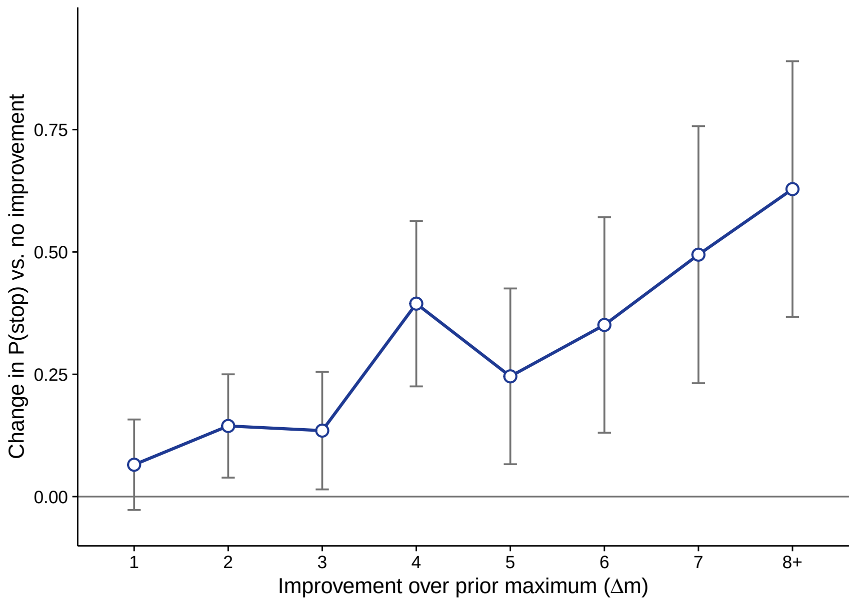
Sample: the 392 participants who opened a second box; one observation each.

delta_m = max(0, second value - first-box value). Column (3) absorbs both the first- and second-value fixed effects, so delta_m is identified only by the kink at the point where the second value crosses the first. SE clustered on participant.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A broad set of robustness checks yields similar stopping estimates under nonlinear hazard models, alternative stopping definitions, sample restrictions, household clustering, exclusion of very high values, and additional survey controls (Appendix Table A.3). Controlling for the expected gain from another opening also leaves the estimate essentially

Figure 2: Stopping by the size of the current improvement



Notes: Coefficients on indicators for Δm (omitted category: no improvement), with step, current-value, prior-maximum, and first-box fixed effects. The 8+ bin combines $\Delta m = 8$ and 9 due to the low number of observations; Figure A.1 in the appendix separates them. Whiskers are 95% confidence intervals with standard errors clustered on participant; $N = 1{,}489$.

unchanged (Appendix Table A.6). Estimates by opening position are strongest at the second and third openings and remain positive, though less precise, later in the search path (Appendix Table A.7).

Result 1. *Search is more likely to end when the current opening improves on the highest value seen so far.*

4.2 Observed evidence and final reports

We next ask whether favorable evidence encountered at the end of the search is reflected in the reported value. To capture how favorable the final opening is, we define *Final improvement* as the amount by which the value revealed in the final opening exceeds the highest value seen before that opening. It is zero if the final opening does not set a new maximum.[8] Table 4 shows that larger final improvements are associated with a higher

[8]For participants who search beyond the first box, *Final improvement* is $\max\{0, x_{iT} - m_{i,T-1}\}$, where T denotes the final opening. It is zero for participants who stop after the first box.

probability of reporting above the truth. The association remains after holding the first-box value and other search-path features fixed (Column 2).[9] Restricting the sample to participants who search beyond the first box increases the coefficient from 0.045 to 0.052 (Column 3). Thus, among participants with the same first-box value and similar search outcomes, reporting above the truth is more common when favorable evidence arrives at the end of the search. Excluding first-box values of 8 and 9 leaves this pattern intact (Appendix Table A.3).

As an additional check, we use the randomized value revealed at the second opening to compare its effect on stopping and subsequent reporting. The results, reported in Appendix Table A.8, show that when the second value exceeds the truth, participants are about 20 percentage points more likely to stop immediately. By contrast, the effect on whether they eventually report above the truth is smaller and imprecisely estimated. This provides further evidence that favorable values affect the decision to stop, while their effect on the overall incidence of higher reports is less clear.

The previous analysis shows that favorable evidence at the end of the search is associated with a greater likelihood of reporting above the truth. We now turn to a different question. When participants do report above the truth, which value do they choose? Appendix Table A.10 shows that *Final improvement* predicts reports within the observed range, but not reports above the highest value observed. Figure 3 shows how closely these reports track the values seen during the search. It plots the difference between the report and the truth, $r - x_1$, against the difference between the highest observed value and the truth, $m_T - x_1$. Most observations lie on the 45-degree line, where the report equals the highest value observed. Fewer lie below it, and only six lie above it.

The point in the search at which the reported value first appears provides further evidence linking stopping and reporting. Among the 96 participants who report above the truth after continued search, 67 report the value revealed at their final opening. For 62 participants, this is also the first time that the reported value appears during the search. To assess whether this could arise simply from the set of values participants happened to observe, we randomly reorder those same values within each search path. Under these random orderings, the corresponding counts fall to 45.3 and 43.2 on average ($p < 0.001$

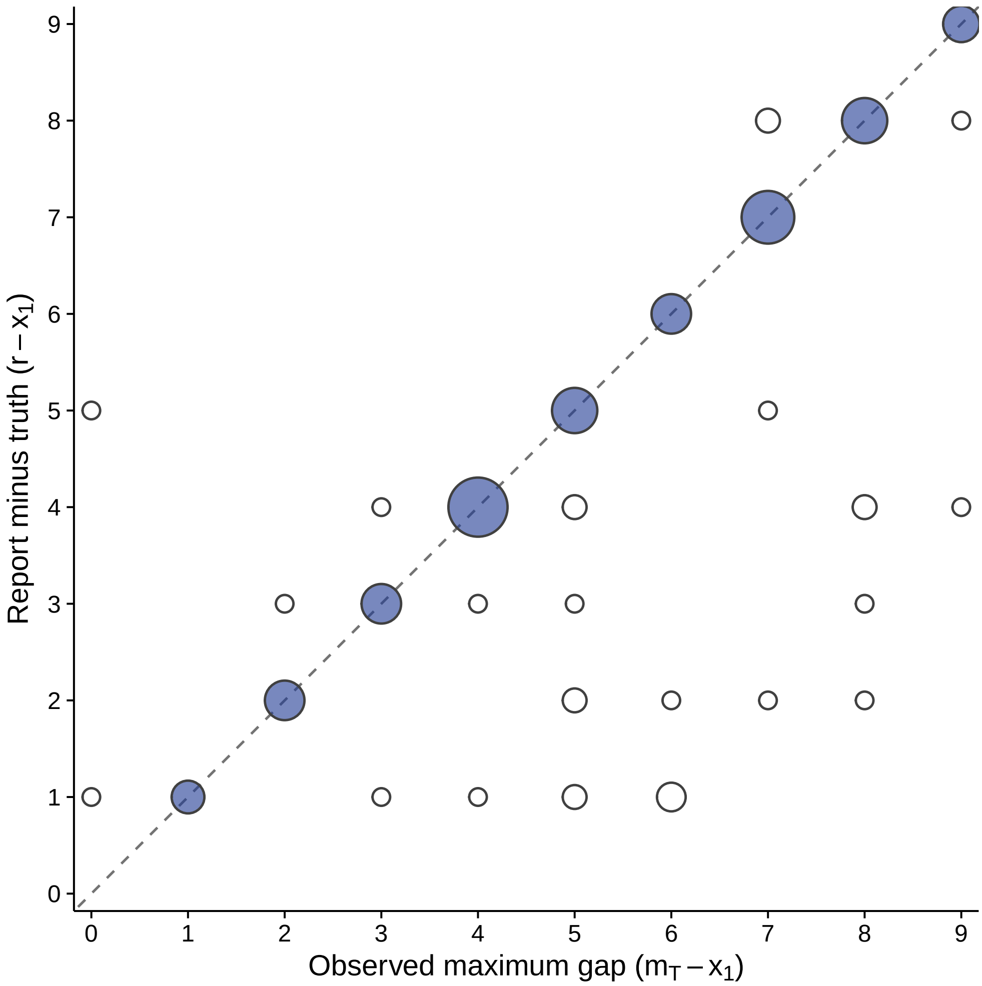
[9] First-box fixed effects hold constant the truth participants were instructed to report. The search indicator distinguishes participants who stop immediately from those who search but do not improve.

Table 4: Terminal favorable evidence and misreporting

	Dependent variable: *Misreport*		
	(1)	(2)	(3)
Final improvement	0.078***	0.045***	0.052***
	(0.010)	(0.013)	(0.013)
Max gap		0.016*	-0.019
		(0.008)	(0.017)
Boxes opened		0.005	0.018**
		(0.008)	(0.009)
Searched		0.065	
		(0.043)	
Observations	1283	1283	392
R^2	0.164	0.228	0.148
Sample	All	All	Searchers
First-box FE	No	Yes	Yes
Outcome mean	0.086	0.086	0.245

Notes: Participant-level LPMs; outcome is upward misreporting. Final improvement is the last opening's improvement in the running maximum and is zero without additional search. Column (2) includes first-box fixed effects and controls for whether the participant searched; Column (3) drops participants who did not open a second box. The estimates are descriptive path decompositions: final improvement, search length, and the observed maximum are jointly generated by the participant's search and stopping decisions. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 3: Observed evidence and reports above the truth after search



Notes: Sample consists of participants who reported above the truth after additional search. Marker area is proportional to the cell count. Navy-filled circles denote reports equal to the observed maximum. The horizontal axis is $m_T - x_1$; the vertical axis is $r - x_1$.

for both; Appendix Table A.9). Thus, the reported value appears at the end of the search substantially more often than would be expected from the observed values alone.

One possible explanation is that some participants mistakenly report the last value they see rather than the first. If this were a neutral instruction error, it should occur whether the final value is higher or lower than the truth. Instead, among all 392 searchers, 67 of 262 participants (25.6%) report the final value when it exceeds the truth, compared with only 7 of 130 (5.4%) when it is lower (Fisher exact $p < 0.001$). Table 5 reports this comparison together with the remaining association between *Final improvement* and reporting.

Table 5: Terminal-value matching and subsequent reports

	Reports terminal value	Reports a post-first opened value	Upward report, excluding terminal-value reports
Terminal value exceeds first	0.177***	0.161***	
	(0.038)	(0.057)	
Final improvement			0.002
			(0.009)
First-box fixed effects	Yes	Yes	Yes
Path controls	No	No	Yes
Observations	392	392	318

Notes: Participant-level OLS among the 392 participants who searched beyond the first box. Raw terminal-value matching is 67/262 (25.6%) when the terminal value exceeds the first-box value and 7/130 (5.4%) when it is below it (Fisher exact $p<0.001$). Columns 1–2 include first-box fixed effects. Column 3 excludes all 74 reports equal to the terminal value and additionally controls for the observed maximum gap and number of boxes opened. Robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Holding the truth fixed, a final value above the truth increases the probability of reporting that final value by 17.7 percentage points. It also increases the probability of reporting any value seen after the first opening by 16.1 percentage points. The same asymmetry appears at the second opening. Participants report the second value in 16.7% of cases when it exceeds the truth, compared with 4.0% when it is lower. This pattern is difficult to reconcile with a neutral mistake of simply reporting the last value seen. Moreover, when we exclude the 74 searchers who report their final value, the coefficient on *Final improvement* falls from 0.052 to 0.002 ($p = 0.861$). Thus, the association between final improvement and reporting above the truth is driven almost entirely by cases in which participants report the value revealed at the end of the search.

Result 2. *Values revealed at the end of the search are disproportionately likely to be reported when they exceed the truth.*

The association between final improvement and reporting above the truth is therefore driven almost entirely by participants who report the value revealed at the end of the search. This pattern alone does not tell us why they do so. One possibility is that high values are simply more salient or easier to remember. Indeed, a simple recall rule that gives greater weight to high values can reproduce both the number of reports above the truth and their concentration at the highest observed value (Appendix Table A.12). However, such a memory-based explanation cannot account for Result 1. It does not explain why participants are more likely to stop searching immediately after a favorable opening, holding the revealed value fixed. The private dice task also provides little evidence that the main search and reporting patterns reflect a general tendency to report high values (Appendix Table A.15). Taken together, the stopping and reporting results show that favorable values relate both to when participants stop searching and to what they ultimately report.

5 Conclusion

This paper studies how discretionary search relates to dishonest reporting. In our survey experiment, participants faced ten boxes containing the values 0 to 9 in random order. They could open as many boxes as they wished, but were instructed to report the value in the first box they opened; higher reports generated higher payments whether or not they were truthful. The first opening therefore fixed the instructed report, while subsequent search could only change the values participants had seen. Because the software recorded every opening, we observe the search path, the stopping decision, and the final report.

Search responds systematically to payoff incentives and to favorable evidence. Participants are less likely to begin searching when the randomized first-box value is already high. Conditional on searching, they are more likely to stop when the current opening improves on every previously observed value, even holding the revealed value fixed, and this response is concentrated in large improvements. At the randomized second opening, a second value above the truth strongly affects immediate stopping but has no clear effect on subsequent misreporting.

Reporting is closely synchronized with stopping. Among participants who search and subsequently misreport upward, reports usually remain within the observed range and often equal its maximum. Favorable terminal values are disproportionately selected as the report itself: terminal matching is much more common when it raises rather than lowers payment, and it occurs more often than expected when the same realized values are randomly reordered. This terminal matching accounts for the association between final improvement and upward misreporting, making it the core report-side result.

These stopping and reporting patterns are more informative together than either is alone. Bunching at observed maxima could arise from memory or salience at the reporting stage, while stopping after favorable values could reflect a search rule unrelated to reporting. In the data, favorable evidence is unusually likely to arrive when search ends, and subsequent misreports are unusually likely to select the value revealed at that stopping point. We call this joint pattern *path-anchored reporting.*

The broader implication is that distortion may be assembled through the search process before a report is chosen. Managers, analysts, researchers, and other decision makers often decide which outcomes to inspect, which comparisons to make, and when to stop before communicating a conclusion. Even when search cannot alter the underlying fact, the realized path determines which evidence is available when the report is made. Final statements alone may therefore conceal an important part of how apparently evidence-based claims were assembled. Search histories, query logs, specification sequences, and stopping times can reveal this process and motivate attention to the rules governing evidence acquisition, not only to the accuracy of the final statement.

The experiment does not distinguish whether favorable evidence makes a false report easier to justify or whether participants simply stop once a value satisfies an aspiration level. Future work can separate these mechanisms by varying whether search is chosen or imposed, whether encountered values may legitimately be cited, or whether the search path is visible to others. The central result does not depend on resolving that distinction: even when later information cannot change what is true, realized search paths predict when search ends and which encountered value is subsequently reported.

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A Additional tables

Table A.1: Individual characteristics

	Mean	SD	Min	Max	N
Female	0.527	0.499	0	1	1318
Age	54.627	18.554	17	95	1318
Education (years)	12.520	2.758	7	18	1279
Net household income (1,000 EUR/month)	3.242	2.124	0	20	1262
Full-time employed (dummy)	0.308	0.462	0	1	1318
Lived in East Germany in 1989	0.233	0.423	0	1	1316
Religious (dummy)	0.613	0.487	0	1	850
Risk willingness [0–10]	5.042	2.252	0	10	1318
Patience [0–10]	5.874	2.537	0	10	1228
Trust in people [1–4]	2.220	0.646	1	4	1225
CRT correct answers (0–3)	0.911	1.029	0	3	933
Openness (Big Five)	4.782	1.155	1	7	1247
Conscientiousness (Big Five)	5.720	0.921	2	7	1246
Extraversion (Big Five)	4.959	1.121	1	7	1247
Agreeableness (Big Five)	5.463	0.950	2	7	1248
Neuroticism (Big Five)	3.785	1.263	1	7	1248
Observations	1318				

Notes: SOEP module sample. Net household income is measured in thousand euros per month.

Table A.2: Randomized first-box value and subsequent search

	Linear first-box value		First-box value 8 or 9	
	Search	Boxes opened	Search	Boxes opened
First-box value	-0.025	-0.060		
	(0.004)	(0.022)		
First-box value 8 or 9			-0.094	-0.177
			(0.029)	(0.154)
Observations	1,283	1,283	1,283	1,283

Notes: Participant-level OLS. *Search* equals one if the participant opens a second box; *Boxes opened* is the number of distinct boxes opened. The first two columns use the randomized first-box value linearly. The last two compare first-box values 8–9 with values 0–7. Robust standard errors in parentheses.

Table A.3: Selected robustness checks for stopping and reporting

Outcome	Specification	Estimate	SE	Observations
Panel A: Current improvement and stopping				
Stop	Baseline OLS	0.067***	0.012	1,489
Stop	Logit average marginal effect	0.062***	0.013	1,446
Stop	Complementary-log-log average marginal effect	0.060***	0.013	1,446
Stop	Final click must reveal new information	0.075***	0.014	1,027
Stop	Terminal revisits treated as censored	0.059***	0.011	1,489
Stop	Mechanically forced terminal openings excluded	0.066***	0.012	1,446
Stop	Participants who ever revisit excluded	0.084***	0.016	843
Stop	Standard errors clustered by household	0.067***	0.012	1,489
Stop	First-box values 8 and 9 excluded	0.069***	0.013	1,207
Stop	Current values 8 and 9 excluded	0.048***	0.017	1,161
Stop	Demographic controls added	0.066***	0.012	1,489
Stop	Usual survey covariates added	0.068***	0.012	1,489
Panel B: Final improvement and reporting				
Upward misreport	First-box values 8 and 9 excluded	0.050***	0.014	1,007

Notes: Each row reports a separate specification. Panel A reports the coefficient on *Current improvement*; Panel B reports the coefficient on *Final improvement*. Unless stated otherwise, the stopping specifications are opening-level OLS models with opening-position, current-value, prior-maximum, and first-box fixed effects and standard errors clustered by participant. The final-click restriction retains paths whose literal final click opens a previously unseen box; the censoring specification assigns no terminal event to paths ending in a revisit. The survey-control rows add demographics and then political orientation, risk willingness, patience, trust, and the Big Five, with missing values mean-imputed and accompanied by missingness indicators. Panel B is the fully adjusted participant-level specification from Table 4, restricted to first-box values below 8, with robust standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Current and accumulated gains. To distinguish whether stopping responds to the arrival of a payoff gain or simply to the best opportunity accumulated during search, Table A.4 compares two measures. *Current gap*, $\max\{0, x_{ik} - x_{i1}\}$, measures how far the value revealed at the current opening lies above the truth, while *Highest-value gap*, $m_{ik} - x_{i1}$, measures how far the best value observed so far lies above the truth.[10]

Table A.4: Current and accumulated gains relative to the first-box value

	Dependent variable: *Stop*			
	(1)	(2)	(3)	(4)
Current gap	0.025**		0.015	
	(0.012)		(0.013)	
Highest-value gap		0.036***		0.031***
		(0.008)		(0.009)
Misreported			-0.061*	-0.042
			(0.033)	(0.041)
Misreported × *Current gap*			0.030**	
			(0.012)	
Misreported × *Highest-value gap*				0.016
				(0.010)
Observations	1489	1489	1489	1489
R^2	0.146	0.154	0.151	0.155
FE	Yes	Yes	Yes	Yes

Notes: Opening-level OLS on post-first openings. Current gap is $\max\{0, x_{ik} - x_{i1}\}$; highest-value gap is $m_{ik} - x_{i1}$. All columns include opening-position, current-value, and first-box fixed effects. Standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Both measures are associated with a higher probability of stopping. The interaction with *Current gap* is clearly positive on paths that eventually end in an upward misreport, whereas the interaction with *Highest-value gap* is smaller and imprecisely estimated. Be-

[10]When the running maximum changes, the highest-value gap equals $(m_{i,k-1} - x_{i1}) + \Delta m_{ik}$.

cause the two interaction effects are not statistically distinguishable, we do not draw a sharp distinction between them. The joint specification in Table A.5 shows the same qualitative pattern.

Table A.5: Timing versus level: horse race

	Dependent variable: *Stop*	
	(1)	(2)
Current gap	0.009	-0.003
	(0.013)	(0.014)
Highest-value gap	0.033***	0.036***
	(0.009)	(0.010)
Misreported		-0.035
		(0.040)
Misreported × *Current gap*		0.036**
		(0.016)
Misreported × *Highest-value gap*		-0.010
		(0.014)
Observations	1489	1489
R^2	0.154	0.159
FE	Yes	Yes

Notes: Opening-level OLS on post-first openings. *Current gap* ($\max\{0, x_{ik} - x_{i1}\}$) and *Highest-value gap* ($m_{ik} - x_{i1}$) are entered jointly. Column (2) adds both interactions with eventual misreporting. A Wald test of equality of the two interaction slopes in Column (2) is reported in the log (`HORSE RACE interaction difference`). All columns include opening-position, current-value, and first-box fixed effects. Standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.6: Stopping and the continuation value of search

	Dependent variable: *Stop*	
	(1)	(2)
Current improvement	0.045***	0.045***
	(0.010)	(0.010)
Expected gain, next opening		0.035
		(0.088)
Observations	1489	1489
R^2	0.168	0.168

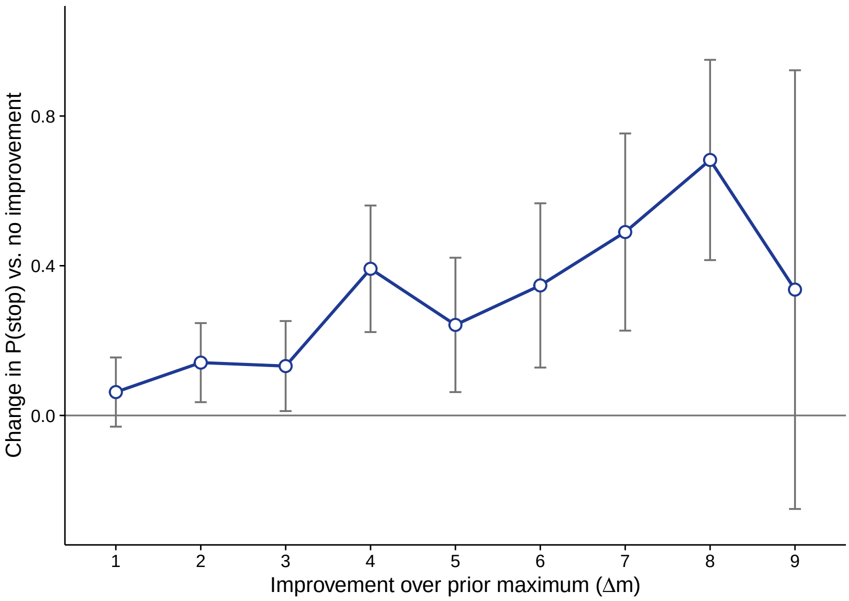
Notes: Opening-level OLS on the sample of Table 2. *Expected gain, next opening* is the exact expected increment in the running maximum from one additional opening, computed from the participant's observed set and the known 0–9 support. Both columns absorb opening-position, current-value, *current*-running-maximum, and first-box fixed effects; conditioning on the current maximum rather than the prior maximum is more demanding, which lowers the *Current improvement* coefficient relative to Table 2. Standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.7: Stopping response to the current improvement, by opening position

	(1)	(2)	(3)	(4)
	Opening 2	Opening 3	Openings 4-5	Openings 6+
Current improvement	0.073***	0.099***	0.042	0.115
	(0.021)	(0.033)	(0.042)	(0.077)
Observations	392	309	391	396
R^2	0.169	0.188	0.127	0.071

Notes: Opening-level OLS on post-first openings. Each column uses openings at the indicated depth and absorbs current-value, prior-maximum, and first-box fixed effects. Columns (3)–(4) pool later openings. Standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A.1: Stopping by the exact size of the current improvement



Notes: Coefficients on indicators for each exact size of the current improvement, Δm (omitted category: no improvement), with step, current-value, prior-maximum, and first-box fixed effects. Whiskers are 95% confidence intervals with standard errors clustered on participant. Improvement sizes 1–9 contain 155, 128, 96, 52, 40, 28, 20, 14, and 3 observations, respectively; $N = 1{,}489$.

Table A.8: The randomized second draw: stopping and eventual reporting

	Stop at second	Final report	Upward report	Report minus truth
Second value	0.038	0.059	0.003	0.059
	(0.007)	(0.042)	(0.008)	(0.042)
Randomization p-value	(<0.001)	0.145	0.640	0.145
Second value exceeds first	0.200	0.502	0.067	0.502
	(0.048)	(0.233)	(0.055)	(0.233)
Randomization p-value	(<0.001)	0.062	0.241	0.062
Improvement size	0.073	0.087	0.005	0.087
	(0.011)	(0.078)	(0.013)	(0.078)
Randomization p-value	(<0.001)	0.237	0.705	0.237
First-box fixed effects	Yes	Yes	Yes	Yes
Observations	392	392	392	392

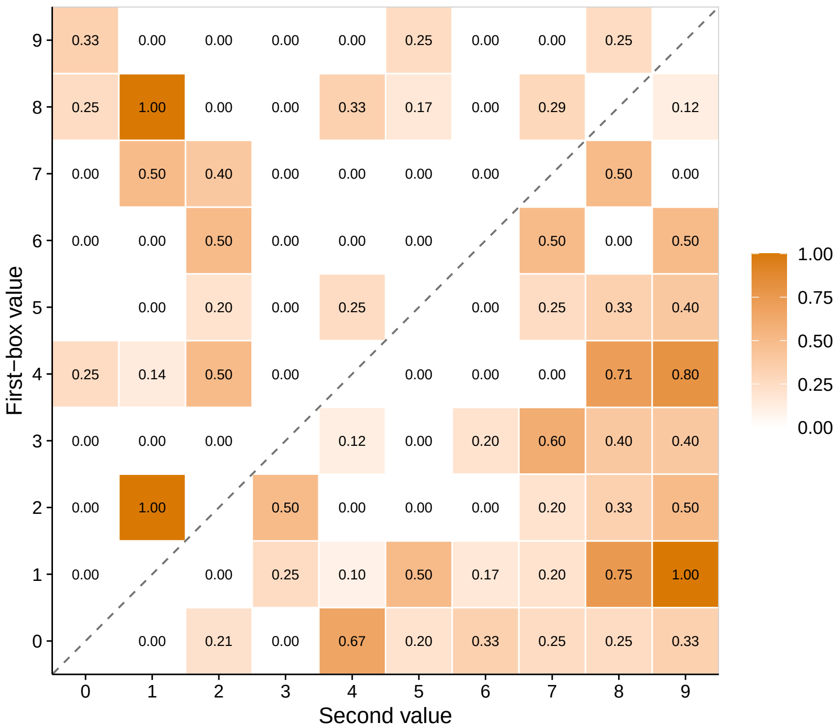
Notes: Each column is a separate participant-level OLS regression among participants who opened a second box. *Second value exceeds first* is an indicator; improvement size is the positive part of the second value minus the first. Robust standard errors are in parentheses. Two-sided randomization p-values use 100,000 draws from the known conditional assignment mechanism and the coefficient as the test statistic.

Table A.9: Report-arrival synchronization after search and misreporting

	Observed count	Share (%) of 96	Random-order expectation	Exact p-value
Final opening equals report	67	69.8	45.27	< 0.001
Final opening first makes report feasible	62	64.6	43.20	< 0.001

Notes: Sample: the 96 participants who misreported after opening at least one additional box. The benchmark holds each participant's first-box value, actual set of post-first values, report, and number of openings fixed, and randomizes only the order of the post-first values. The first row counts participants whose final opening reveals exactly the value they later report; the second counts participants for whom the terminal opening is the first point at which the running maximum reaches the eventual report. p-values are exact upper-tail probabilities from the Poisson-binomial distribution implied by the participant-specific benchmark probabilities.

Figure A.2: Stopping at the second opening, by first-box value and second value



Notes: Each cell reports the share of the 392 second-opening searchers with that (first-box value, second value) pair who stop searching; darker shading indicates a higher stopping share. Improvements are the cells in which the second value exceeds the first-box value. The 89 occupied cells contain 1–14 observations (mean 4.4), so individual cell rates are descriptive; Table 3 provides the pooled estimates.

Reports inside and above the observed range. Table A.10 separates bounded misreports, $x_1 < r \leq m_T$, from reports above the observed range, $r > m_T$. *Final improvement* predicts bounded misreports (0.045) but not reports above the observed maximum (0.001); the difference between the two coefficients is 0.044 ($p = 0.001$).[11] This decomposition supports the path interpretation but is secondary to the terminal-matching result in Table 5.

Table A.10: Terminal favorable evidence and the type of misreport

	(1) Any upward	(2) Bounded	(3) Unbounded
Final improvement	0.045***	0.045***	0.001
	(0.013)	(0.013)	(0.003)
Max gap	0.016*	0.023***	-0.007**
	(0.008)	(0.008)	(0.003)
Boxes opened	0.005	0.006	-0.001
	(0.008)	(0.008)	(0.002)
Searched	0.065	0.045	0.021
	(0.043)	(0.040)	(0.017)
Observations	1283	1283	1283
R^2	0.228	0.271	0.017
Outcome mean	0.086	0.070	0.016

Notes: Participant-level LPMs with the controls of Table 4, Column (2), including first-box fixed effects. *Bounded* marks upward misreports at or below the maximum value observed ($x_1 < r \leq m_T$); *Unbounded* marks reports above it ($r > m_T$), including the misreports entered without additional search. The difference between the bounded and unbounded *Final improvement* slopes is 0.044 (SE 0.013), $p = 0.001$. Because report type and final improvement are both produced by the path, the columns are descriptive decompositions rather than causal effects of terminal evidence. Robust standard errors in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

[11] Among the 392 participants who search beyond the first box, the corresponding slopes are 0.050 and 0.002; their difference is 0.048 ($p < 0.001$).

Table A.11: Randomization benchmarks for terminal favorable evidence

	Observed	Null	p-value
Panel A: Redrawing paths conditional on first box and search length			
All: terminal Δm	1.582	0.929	< 0.001
All: terminal new max	0.462	0.314	< 0.001
Upward: terminal Δm	2.750	1.486	< 0.001
Upward: terminal new max	0.708	0.435	< 0.001
Panel B: Permuting the order of observed post-first values			
All: terminal Δm	1.582	1.257	< 0.001
All: terminal new max	0.462	0.349	< 0.001
Upward: terminal Δm	2.750	2.161	< 0.001
Upward: terminal new max	0.708	0.514	< 0.001

Notes: Upward denotes participants who reported above the first-box value. Panel A holds the first-box value and search length fixed, redraws additional distinct values without replacement, and keeps the terminal index fixed. Panel B holds each participant's observed post-first values fixed, randomly permutes their order, and keeps the terminal index fixed. These are descriptive ordering benchmarks that condition on realized path features, not randomization tests of a fully specified stopping model. The table reports upper-tail simulation p-values from 2000 draws; values below 0.001 are displayed as < 0.001.

Table A.12: Reporting-side recall benchmarks, including value-salience recall

Recall model	E[upward]	p	E[exact max]	p
Uniform	82.6	< 0.001	53.9	< 0.001
Linear recency	86.0	0.002	58.9	0.011
Extremeness (value-weighted)	96.3	0.632	68.5	0.607
Maximum-status boost	90.3	0.046	70.5	0.770
Observed	96		68	

Notes: Sample: the 115 participants who searched and reported a value other than the first-box value. Each model draws the recalled report from the participant's own opened values with the stated weights; p-values are upper-tail Poisson-binomial simulations (50,000 draws). Uniform and linear-recency recall are rejected, but recall weighted by the value itself (payoff salience) reproduces the observed upward and exact-maximum counts. The reporting side alone therefore does not reject motivated memory; the stopping result—which a reporting-stage recall error cannot generate—does.

Mechanical reporting benchmarks. Mechanical error rules also fail to reproduce the reporting distribution jointly. Applied to all 392 searchers, always reporting the terminal value predicts 66.8% upward and 33.2% downward reports, compared with 24.5% and 4.8% in the data. Uniform, value-weighted, and recency-weighted recall likewise predict far too little truth telling and too many downward reports. These comparisons do not rule out every form of confusion, but they show that simple report-the-last, report-the-maximum, and recall rules cannot jointly explain the truth, upward, downward, maximum, and terminal moments.

Table A.13: Observed reporting and mechanical error benchmarks

	Truth	Upward	Downward	Exact maximum	Exact terminal
Observed reports	70.7	24.5	4.8	32.9	18.9
Always report terminal value	0.0	66.8	33.2	46.2	100.0
Uniform opened-value recall	27.6	42.1	30.3	27.6	27.6
Report most recent new maximum	17.6	82.4	0.0	100.0	46.2
Value-weighted recall	22.0	61.4	16.6	48.1	35.7
Recency-weighted recall	13.8	51.8	34.4	32.4	41.5

Notes: Percentages among all 392 participants who opened at least two boxes. Each benchmark is evaluated on every participant's realized set and order of opened values. Uniform recall gives every opened value equal weight; value-weighted recall assigns weight proportional to the value; recency-weighted recall assigns linearly increasing weight to opening position. No benchmark is fitted to the observed moments.

External calibration. Table A.14 compares reports with the *Numbers* treatment of Gneezy et al. (2018), which has the same value support but no additional search. That cross-study comparison also produces much less concentration at participant-specific observed maxima. Because it compares different samples and conditions on upward misreporting, we treat it as descriptive external validation rather than as the primary inferential benchmark.

Table A.14: Bounded reports against an external pathless benchmark

	Observed count	Share (%) of 96	Pathless expectation	Exact p-value
Bounded report ($r \leq m_T$)	90	93.8	53.7 (55.9%)	< 0.001
Exact observed maximum ($r = m_T$)	68	70.8	31.9 (33.3%)	< 0.001

Notes: Sample: the 96 participants who misreport upward after opening at least one additional box. The benchmark assigns each participant a counterfactual report drawn from the empirical report distribution of the 103 participants who misreport upward in the observed *Numbers* treatment of Gneezy et al. (2018), conditional on having observed the same first-box value (payoffs 1–10 mapped to the 0–9 support). That environment has no search path: participants click a single box and observe one value under comparable stakes. Expectations sum the participant-specific benchmark probabilities that the counterfactual report is weakly below (row 1) or exactly equal to (row 2) the participant's observed maximum. p-values are exact upper-tail probabilities from the corresponding Poisson-binomial distributions, treating the external empirical distribution as fixed. The comparison is descriptive because the two studies use different participant populations.

Dice-task comparison. The dice task provides a complementary measure of participants' tendency to submit high reports when their true outcome is private. Because the first roll is unobserved, we cannot identify individual misreporting. We therefore measure *Dice excess* as the reported value minus the expected value of a truthful report (4.5) and use it only as a proxy for high reporting in the dice task. Task order was randomized, and the main box- and dice-task outcomes do not differ detectably by order.

We first ask whether a general tendency to submit high reports explains the relationship between terminal favorable evidence and box-task misreporting. Columns 1–2 of Table A.15 add *Dice excess* to the participant-level misreporting regressions. The coefficient on *Final improvement* is essentially unchanged, while the association between *Dice excess* and box-task misreporting is small.

We then examine whether high reporting in the dice task is related to search behavior itself. *Dice excess* does not predict whether participants search or how many boxes they open, although it is positively associated with the favorable evidence available at the end of search (Table A.15, Columns 5–7). At the opening level, including *Dice excess* leaves the relationship between *Current improvement* and stopping essentially unchanged (Column 3), while its interaction with *Current improvement* is positive but modest (Column 4). Thus, high reporting across tasks is related to some box-task outcomes, but it does not account for the main search and reporting relationships.

Table A.15: Cross-task robustness and box-task behavior

	Person level		Opening level		Participant-level behavior		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Misreport	Misreport	Stop	Stop	Searched	Boxes opened	Final improvement
Final improvement	0.043***	0.044***					
	(0.013)	(0.013)					
Dice excess	0.005*	0.005**	0.002	-0.003	0.004	0.000	0.037***
	(0.002)	(0.002)	(0.004)	(0.004)	(0.005)	(0.023)	(0.014)
Current improvement			0.067***	0.061***			
			(0.012)	(0.012)			
Current improvement × Dice excess				0.005**			
				(0.002)			
Observations	1283	1283	1489	1489	1283	1283	1283
R^2	0.222	0.231	0.175	0.178	0.049	0.025	0.100
FE	No	Yes	Full	Full	First-box	First-box	First-box
Outcome mean	0.086	0.086	0.263	0.263	0.306	2.161	0.483

Notes: Columns (1)–(2) are participant-level LPMs for upward misreporting and additionally control for max gap, boxes opened, and search. Columns (3)–(4) are opening-level OLS stopping regressions on post-first openings. Columns (5)–(7) are participant-level OLS models relating dice excess to whether participants search, the number of boxes opened, and final improvement; they include task order and first-box fixed effects. Dice excess is the reported dice value minus 4.5. Participant-level standard errors are robust; opening-level standard errors are clustered by participant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B An illustrative reporting formalization

This appendix gives one possible microfoundation for the reporting patterns in the paper. It is an illustrative conditional reporting problem, not an estimated model, and it does not derive the opening-level stopping relationship.

Fix the state when a participant stops after k openings. The truthful report is x_1, the largest observed value is m_k, and the current improvement is Δm_k. The participant chooses a report r from the finite support $X = \{0, \ldots, 9\}$. Reporting r yields material payoff r. Let

$$J_k \equiv J(x_1, m_k, \Delta m_k) \geq 1$$

denote an evidence factor: a bounded report is less costly when J_k is larger. The observed bound and the convex cost below generate reports at and below the observed maximum. To represent terminal salience, assume that J_k is weakly increasing in Δm_k. This assumption motivates the timing contrast in the empirical analysis; it is not separately identified by the model or the data.

For $\kappa > 0$, consider the conditional reporting utility

$$U_k(r) = r - \begin{cases} \dfrac{\kappa[(r - x_1)_+]^2}{J_k}, & r \leq m_k, \\[2ex] \kappa[(r - x_1)_+]^2, & r > m_k. \end{cases} \tag{1}$$

Reports no greater than the observed maximum receive the evidence discount; reports above it do not. The quadratic form is convenient rather than substantive. Its relevant features are that the cost rises with the size of the lie and that the observed maximum defines a range of reports receiving the discount. A downward report is dominated by the truth: both carry zero lying cost, and the truth pays more. We therefore treat the 21 downward reports as non-strategic error outside this account. Because J_k enters only the bounded branch, a terminal update does not directly raise the utility of an above-observed report, although its observed share can change as the two regions compete.

Define the bounded and above-observed reporting regions as

$$X_k^B \equiv \{r \in X : x_1 \leq r \leq m_k\}, \qquad X_k^A \equiv \{r \in X : r > m_k\}.$$

Proposition B.1 (Conditional reporting rule)**.** *Within the bounded region, the continuous target is*

$$\widehat{r}_k^B = \min\left\{m_k,\ x_1 + \frac{J_k}{2\kappa}\right\}.$$

The maximizers over X_k^B are the feasible support point or points closest to $\widehat{r}_k^B$. Within the above-observed branch, the unconstrained quadratic target is

$$\widetilde{r}_k^A = x_1 + \frac{1}{2\kappa},$$

and the maximizers over a nonempty X_k^A are its feasible support point or points closest to $\widetilde{r}_k^A$. The global optimum compares the best value of $U_k(r)$ in the two regions.

Proof. For any $q > 0$ and $r \geq x_1$, completing the square gives

$$r - \kappa\frac{(r - x_1)^2}{q} = x_1 + \frac{q}{4\kappa} - \frac{\kappa}{q}\left(r - x_1 - \frac{q}{2\kappa}\right)^2.$$

Set $q = J_k$ on X_k^B and impose the upper bound m_k; set $q = 1$ on X_k^A. Projection onto the finite support and comparison across regions give the result. □

The bounded target rises with the evidence factor. In the corresponding continuous problem,

$$\widehat{r}_k^B = m_k \quad \Longleftrightarrow \quad J_k \geq 2\kappa(m_k - x_1).$$

Below this threshold, the participant optimally reports below the observed maximum; above it, the target equals that maximum. With finite support, rounding makes the threshold sufficient but not necessary for an exact-maximum report. The same formulation also permits an above-observed report when the best utility on X_k^A exceeds the best bounded alternative.

The formalization therefore rationalizes bunching at observed maxima without requiring every report to equal the maximum. It does not imply that a favorable improvement causes stopping. Holding continuation value fixed, a larger evidence factor weakly raises the best stopping-and-reporting payoff,[12] but a favorable opening may also change continuation value. An aspiration rule can produce the same stopping pattern without the reporting cost in (1). We therefore treat the stopping relationship as an empirical association and the formalization as one possible interpretation.

[12] For each fixed $r \leq m_k$, increasing J_k weakly lowers the cost in (1); utilities for $r > m_k$ are unchanged. The maximum over $r \in X$ therefore weakly rises.

C Instructions

The text below reproduces the participant instructions verbatim. Interviewer instructions appear in *italics* and were not shown to participants. Task order was randomized; headings follow the module documentation.

Intro

Now I would like to invite you to participate in a short and simple game. The task consists of two parts. What the two parts are about, you will learn immediately before. In both parts, you have a chance to win real money. But only one of the two parts will be selected for the final payout. You will flip a coin for it. If the coin shows "tails" after the toss, you will get the result for part 1. If the coin shows "heads", you will get the result for part 2. All information from the game, like all other information from this interview, will be evaluated anonymously only and will not be associated with your name. This is the end of the introduction.

Interviewer: Please do not inform the target person in advance about the content of the two games! As the interviewer, make sure that the target person does not see the dice cup and dice before the dice game begins. Please do not give the dice cup and dice to the target person until the target person asks you for it!

Q490

Would you like to play the game?

Interviewer: Before starting the module: As the interviewer, please try to make sure that no other interviewee is within earshot. As the interviewer, also try to leave the room or turn away so that the target person can answer the questions alone. It is important that the target person can complete the following tasks in complete privacy. If the target person agrees to play the game, please hand over the laptop to the target person for self-completion. After the experiment, the target person will be asked to hand over the laptop to you and continue the interview as regular.

- Yes
- No

Q491

Interviewer: Please hand over the laptop to the interviewee for self-completion.

Q493: Dice task

The interviewer has just given you a 10-sided die and a dice cup. Now please roll the dice as many times as you like using the dice cup. Caution: Make sure that the interviewer does not see the result of your roll. Please remember the result of your first roll and enter it on the screen. How much you can earn in this part, if this part of the task is selected for payout, depends on the number you write down here. A higher number means a higher payout. You can see the exact payout for each reported number in the table below. The table will remain on the screen until you enter your number.

Note: Please have your interviewer hand you the dice cup and dice. Please make sure that no one can observe you playing the game.

[Participants saw a table reflecting the payoffs for all values they could report]

Q494: Box task

In this part, you will see 10 black boxes on your screen. There are numbers behind the black boxes. The numbers behind the black boxes are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The numbers are in random order, so the first box does not necessarily contain 1, the second does not necessarily contain 2, and so on. As soon as you click on a black box, the number hidden behind it is displayed. You can click on as many black boxes as you like. The laptop records how many and which boxes you click. Please remember the number behind the first black box you clicked and enter it on the screen. How much you earn in this part, if this part of the task is selected for payout, depends on the number you enter. A higher number means a higher payout. You can see the exact payout for each number reported in the table below. The table will remain on the screen until you enter your number.

[Participants saw a table reflecting the payoffs for all values they could report]

Q495

You have reached the end of the task. Please hand the laptop back to your interviewer(s) and continue the survey as usual.

Q496_0

Now we will determine whether the first or second part of this task will be paid out. To do this, I ask you to flip a coin in a clearly visible position so that I can note down the result. Ask the target person to toss a coin in plain sight. Note whether "tails" or "heads" is on top.

Interviewer: Ask the target person to toss a coin in plain sight. Note whether "tails" or "heads" is on top.

Q496

As you can see, [PROG: Show result from IZAHLAUSZ01] is on top. Therefore, you will now receive the payoff from [PROG: please show: Part 1 (if IZAHLAUSZ01=1); Part 2 (if IZAHLAUSZ01=2)]. You have indicated that in [PROG: please show: Part 1 (if IZAHLAUSZ01=1); Part 2 (if IZAHLAUSZ01=2)] the first number was [PROG: please enter the number the participant entered based on their split group (SPLITZAHL) in either Part 1 or Part 2]. Therefore, you will now receive euros from me [PROG: reported number = euro amount].

Interviewer: Please pay the target person the appropriate amount.